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Cooling system

Abstract

A cooling system for cooling a fuel cell on a vehicle includes a radiator, a branch portion connected to an outlet side of the radiator, a confluence portion connected to an inlet side of the radiator, a first passage and a second passage connected in parallel between the confluence portion and the branch portion, a fuel cell and a first pump provided in the first passage, a resistor and a second pump provided in the second passage, and a backflow preventer provided in the second passage. The first passage has no backflow preventer.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION
(1) This application is based on Japanese Patent Application No. 2021-112961 filed on Jul. 7, 2021, the disclosure of which is incorporated herein by reference.
TECHNICAL FIELD
(2) The present disclosure relates to a cooling system for cooling a fuel cell mounted on a vehicle.
BACKGROUND
(3) A cooling system cools a resistor for a brake by blowing air from a blower. The cooling system is installed in a vehicle in which a generator is driven by an internal combustion engine as a power source and the generated power is used to drive an electric motor. The resistor converts the electric power into heat energy when a braking operation is conducted to rotation of wheels mechanically connected to the electric motor.
SUMMARY
(4) According to an aspect of the present disclosure, a cooling system that cools a fuel cell for a vehicle includes: a radiator configured to radiate heat of a coolant, the radiator having a radiator inlet for a coolant to flow in and a radiator outlet for a coolant to flow out; an outlet passage connected to the radiator outlet for a coolant flowing out of the radiator; an inlet passage connected to the radiator inlet for a coolant flowing toward the radiator; a branch portion provided in the

outlet passage for branching into a plurality of passages for a coolant; a confluence portion provided in the inlet passage for merging the plurality of passages into one passage; a first passage and a second passage, as the plurality of passages, connected in parallel with each other between the branch portion and the confluence portion, a coolant flowing from the branch portion toward the confluence portion; a fuel cell provided in the first passage, the fuel cell having a fuel cell inlet for a coolant to flow in and a fuel cell outlet for a coolant to flow out, a coolant flowing inside the fuel cell, the fuel cell generating electric power by an electrochemical reaction of fuel; a first pump provided in the first passage to pump a coolant; a resistor provided in the second passage for a brake, the resistor having a resistor inlet for a coolant to flow in and a resistor outlet for a coolant to flow out, a coolant flowing inside the resistor, the resistor converting an electric power generated by a motor generator into a heat energy by a braking operation to rotation of a wheel mechanically connected to the motor generator; a second pump provided in the second passage to pump a coolant; and a backflow preventer provided in the second passage as a mechanical component different from the second pump, so as to prevent a coolant from flowing inside the resistor in a reverse direction from the resistor outlet to the resistor inlet.
(5) The first passage has no backflow preventer as a mechanical component different from the first pump to prevent a coolant from flowing inside the fuel cell in a reverse direction from the fuel cell outlet to the fuel cell inlet.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic diagram showing a cooling system according to a first embodiment.
(2) FIG. **2** is a schematic diagram showing a cooling system of Comparative Example 1.
(3) FIG. **3** is a schematic diagram showing a cooling system of Comparative Example 2.
(4) FIG. **4** is a schematic diagram showing a cooling system according to a second embodiment.
(5) FIG. **5** is a flowchart of a control process executed by a control device of the second embodiment.
(6) FIG. **6** is a schematic diagram showing a cooling system according to a third embodiment.
(7) FIG. **7** is a flowchart of a control process executed by a control device of the third embodiment.
(8) FIG. **8** is a schematic diagram showing a cooling system according to a fourth embodiment.
(9) FIG. **9** is a flowchart of a control process executed by a control device of the fourth embodiment.
(10) FIG. **10** is a schematic diagram showing a cooling system according to a fifth embodiment.
(11) FIG. **11** is a flowchart of a control process executed by a control device of the fifth embodiment.
(12) FIG. **12** is a schematic diagram showing a cooling system according to a sixth embodiment.
DESCRIPTION OF EMBODIMENTS
(13) To begin with, examples of relevant techniques will be described.
(14) A cooling system that cools a resistor for a brake by blowing air from a blower. The cooling system is installed in a vehicle in which a generator is driven by an internal combustion engine as a power source and the generated power is used to drive an electric motor. The resistor converts the electric power into heat energy when a braking operation is conducted to rotation of wheels mechanically connected to the electric motor.
(15) The present inventors have examined a cooling system for cooling each of a fuel cell mounted on a vehicle and a resistor for a brake by heat exchange using a coolant. The cooling system is called as a first cooling system.
(16) The first cooling system includes: a radiator that radiates heat of a coolant; an outlet passage connected to an outlet of the radiator for the coolant; an inlet passage connected to an inlet of the

radiator for the coolant; a branch portion provided downstream of the outlet passage; a confluence portion provided upstream of the inlet passage; a first passage and a first passage connected in parallel with each other between the branch portion and the confluence portion; a fuel cell and a first pump provided in the first passage; and a resistor for a brake and a second pump provided in the second passage. Each of the first pump and the second pump sends the coolant by rotation of the impeller.

(17) According to the first cooling system, one radiator is used for the combination of the fuel cell and the resistor. The one radiator is used for radiating heat of both of the coolant received from the fuel cell and the coolant received from the resistor. Therefore, the cooling system can be downsized as compared with a case where the cooling system separately includes a radiator for the fuel cell and a radiator for the resistor.

(18) Further, according to the first cooling system, the fuel cell and the resistor are connected in parallel relative to the one radiator. The pumps are provided respectively for the fuel cell and the resistor to pump the coolant. Therefore, it is easy to control the flow rate of the coolant flowing through the fuel cell and the flow rate of the coolant flowing through the resistor to a target flow rate.

(19) However, the frequency of cooling the resistor is lower than the frequency of cooling the fuel cell. Therefore, it may not be necessary to cool the resistor while it is necessary to cool the fuel cell. In this case, the first pump operates and the second pump stops. As a result, while a part of the coolant flowing out of the fuel cell flows through the radiator, the other part of the coolant flowing out of the fuel cell flows backward in the second passage and flows into the fuel cell without flowing through the radiator. The present inventors found that the heat dissipation capacity of the radiator is not sufficiently exerted for cooling the fuel cell.

(20) In view of the above points, it is an object of the present disclosure to provide a cooling system capable of sufficiently exerting the heat dissipation capacity of the radiator for cooling the fuel cell.

(21) According to an aspect of the present disclosure, a cooling system that cools a fuel cell for a vehicle includes: a radiator configured to radiate heat of a coolant, the radiator having a radiator inlet for a coolant to flow in and a radiator outlet for a coolant to flow out; an outlet passage connected to the radiator outlet for a coolant flowing out of the radiator; an inlet passage connected to the radiator inlet for a coolant flowing toward the radiator; a branch portion provided in the outlet passage for branching into a plurality of passages for a coolant; a confluence portion provided in the inlet passage for merging the plurality of passages into one passage; a first passage and a second passage, as the plurality of passages, connected in parallel with each other between the branch portion and the confluence portion, a coolant flowing from the branch portion toward the confluence portion; a fuel cell provided in the first passage, the fuel cell having a fuel cell inlet for a coolant to flow in and a fuel cell outlet for a coolant to flow out, a coolant flowing inside the fuel cell, the fuel cell generating electric power by an electrochemical reaction of fuel; a first pump provided in the first passage to pump a coolant; a resistor provided in the second passage for a brake, the resistor having a resistor inlet for a coolant to flow in and a resistor outlet for a coolant to flow out, a coolant flowing inside the resistor, the resistor converting an electric power generated by a motor generator into a heat energy by a braking operation to rotation of a wheel mechanically connected to the motor generator; a second pump provided in the second passage to pump a coolant; and a backflow preventer provided in the second passage as a mechanical component different from the second pump, so as to prevent a coolant from flowing inside the resistor in a reverse direction from the resistor outlet to the resistor inlet.

(22) The first passage has no backflow preventer as a mechanical component different from the first pump to prevent a coolant from flowing inside the fuel cell in a reverse direction from the fuel cell outlet to the fuel cell inlet.

(23) Accordingly, the backflow preventer is provided in the second passage having the resistor.

Therefore, when the second pump is stopped and the first pump is operating, a part of the coolant flowing out from the fuel cell is prevented from flowing backward in the second passage, such that all the coolant flowing out from the fuel cell can flow through the radiator. As a result, the heat dissipation capacity of the radiator can be fully exerted for cooling the fuel cell.
(24) When it is necessary to cool the resistor but the fuel cell does not need to be cooled, the second pump is operated for cooling the resistor and the first pump is stopped. At this time, if a part of the coolant flowing out of the resistor flows back through the first passage and flows into the resistor without flowing through the radiator, the heat dissipation capacity of the radiator cannot be fully used for cooling the resistor. From the viewpoint of fully exerting the heat dissipation capacity of the radiator for cooling the resistor, it is conceivable to provide a backflow preventer in the first passage for the fuel cell in addition to the second passage for the resistor.
(25) However, the amount of heat radiation required for the resistor is less than the amount of heat radiation required for the fuel cell. Therefore, even if a part of the coolant flowing out of the resistor flows back through the first passage and the heat dissipation capacity of the radiator is not sufficiently exerted for cooling the resistor, the influence is small compared with a case of cooling the fuel cell.
(26) Rather, in case where a backflow preventer is provided in the first passage, the pressure loss of the coolant is increased by the backflow preventer when the coolant flows through the first passage, while the first pump is operated to cool the fuel cell. Therefore, in order to secure the same flow rate of the coolant flowing through the first passage as in the case where the backflow preventer is not provided, it is necessary to increase the rotation speed of the first pump provided in the first passage. This causes decrease in the life of the first pump and deterioration in the fuel efficiency. Further, the number of components needed for the cooling system is increased by providing the backflow preventer. Since the cooling system is mounted on the vehicle, it is better to have few components as the cooling system.
(27) Therefore, according to the first aspect of the present disclosure, the backflow preventer is not provided in the first passage for the fuel cell. As a result, the pressure loss generated in the coolant flowing through the first passage can be reduced as compared with a case where the backflow preventer is provided in the first passage. Further, the number of components needed for the cooling system can be reduced as compared with a case where the backflow preventer is provided in both the first passage and the second passage.
(28) The reference numerals attached to the components and the like indicate an example of correspondence between the components and the like and specific components and the like described in embodiments to be described below.
(29) Hereinafter, embodiments of the present disclosure will be described with reference to the drawings. In the following embodiments, the same or equivalent parts are denoted by the same reference numerals.
First Embodiment
(30) A cooling system **1** of the present embodiment shown in FIG. **1** is mounted on a large commercial vehicle such as a truck, a bus, or a trailer with a fuel cell. A vehicle equipped with a fuel cell is referred to as a fuel cell vehicle. The cooling system **1** cools a fuel cell **2** and a resistor **3** mounted on the large commercial vehicle with a coolant.
(31) The fuel cell **2** generates electric power by an electrochemical reaction of fuel. As the fuel cell **2**, a solid polymer type fuel cell is used. The electric power generated by the fuel cell **2** is used to drive a motor generator (such as an electric motor, not shown) mounted on the fuel cell vehicle. By driving the motor generator, the wheels mechanically connected to the motor generator rotate, and the propulsive force is obtained for driving the fuel cell vehicle.
(32) The resistor **3** is for a brake, and converts the electric power generated by the motor generator into heat energy. The motor generator generates electricity by braking against the rotation of the wheels when the fuel cell vehicle decelerates. When the battery can be charged, the power

generated by the motor generator is charged into the battery. When the battery cannot be charged due to a fully charged state or the like, the electric power generated by the motor generator is supplied to the resistor **3**.

(33) The coolant is a heat exchange medium in a liquid state. The coolant is used in the liquid state when the fuel cell **2** and the resistor **3** are cooled. The coolant does not include refrigerant for a refrigeration cycle, which changes its state between the liquid and the gas during use.

(34) The cooling system **1** includes a coolant circuit **10** through which the coolant circulates. The coolant circuit **10** includes a radiator **11**, an outlet passage **12**, an inlet passage **13**, a branch portion **14**, a confluence portion **15**, a first passage **16**, a second passage **17**, the fuel cell **2**, a first pump **18**, the resistor **3**, a second pump **19**, and a check valve **20**.

(35) The radiator **11** radiates heat of the coolant by a heat exchange between the coolant and air. The radiator **11** has a coolant inlet **11***a* and a coolant outlet **11***b*. The coolant inlet **11***a* is a radiator inlet for the coolant to flow into the radiator **11**. The coolant outlet **11***b* is a radiator outlet for the coolant to flow out from the radiator **11**.

(36) The outlet passage **12** is connected to the coolant outlet **11***b* of the radiator **11**. The coolant flowing out of the coolant outlet **11***b* flows through the outlet passage **12**. The inlet passage **13** is connected to the coolant inlet **11***a* of the radiator **11**. The coolant flows through the inlet passage **13** toward the radiator **11**.

(37) The branch portion **14** is provided downstream of the outlet passage **12** in a flow of the coolant. The branch portion **14** branches the outlet passage **12** for the coolant into plural passages such as a first passage **16** and a second passage **17**. The confluence portion **15** is provided upstream of the inlet passage **13**. The confluence section **15** merges the plural passages into one passage.

(38) The first passage **16** and the second passage **17** are connected in parallel with each other between the branch portion **14** and the confluence portion **15**. The first passage **16** and the second passage **17** are plural passages through which the coolant flows from the branch portion **14** toward the confluence portion **15**. One end of the first passage **16** and one end of the second passage **17** are connected to the branch portion **14**. The other end of the first passage **16** and the other end of the second passage **17** are connected to the confluence portion **15**.

(39) The fuel cell **2** is provided in the middle of the first passage **16**. The fuel cell **2** has a coolant inlet **2***a* and a coolant outlet **2***b*. The coolant inlet **2***a* is a fuel cell inlet for the coolant to flow into the fuel cell **2**. The coolant outlet **2***b* is a fuel cell outlet for the coolant to flow out of the fuel cell **2**. The coolant flows inside the fuel cell **2**. That is, the fuel cell **2** has an internal passage (not shown) formed inside the fuel cell **2**. The coolant flows from the coolant inlet **2***a* toward the coolant outlet **2***b* through the internal passage of the fuel cell **2**. When the coolant flows through the internal passage of the fuel cell **2**, the coolant receives heat from the fuel cell **2**. That is, the fuel cell **2** radiates the heat to the coolant.

(40) The first pump **18** is provided in the middle of the first passage **16**. Specifically, the first pump **18** is provided at a position between the branch portion **14** and the fuel cell **2** in the first passage **16**. That is, the first pump **18** is provided upstream of the fuel cell **2** in a flow of the coolant in the first passage **16**. The first pump **18** has a casing (not shown) and an impeller (not shown). The first pump **18** is a non-volumetric electric pump that sends a coolant by rotating the impeller in the casing.

(41) The resistor **3** is provided in the middle of the second passage **17**. The resistor **3** has a coolant inlet **3***a* and a coolant outlet **3***b*. The coolant inlet **3***a* is a resistor inlet for the coolant to flow into the resistor **3**. The coolant outlet **3***b* is a resistor outlet for the coolant to flow out of the resistor **3**. The coolant flows inside the resistor **3**. That is, the resistor **3** has an internal passage (not shown) inside the resistor **3**. The coolant flows from the coolant inlet **3***a* toward the coolant outlet **3***b* through the internal passage of the resistor **3**. When the coolant flows through the internal passage of the resistor **3**, the coolant receives heat from the resistor **3**. That is, the resistor **3** radiates the heat to the coolant.

(42) The second pump **19** is provided in the middle of the second passage **17**. Specifically, the second pump **19** is provided at a position between the branch portion **14** and the resistor **3** in the second passage **17**. That is, the second pump **19** is provided upstream of the resistor **3** in a flow of the coolant in the second passage **17**. The second pump **19** is a non-volumetric electric pump that sends a coolant by rotating an impeller in a casing, similarly to the first pump **18**.
(43) The check valve **20** is provided in the middle of the second passage **17**. The check valve **20** is a mechanical component different from the second pump **19**, and is a backflow preventer for preventing the backflow of the coolant in the second passage **17**. The backflow of the coolant in the second passage **17** means that the coolant flows in the second passage **17** in the reverse direction from the coolant outlet **3***b* toward the coolant inlet **3***a*, inside the resistor **3**. The check valve **20** is provided at a position between the coolant outlet **3***b* of the resistor **3** and the confluence portion **15** in the second passage **17**. The position where the check valve **20** is provided is not limited to this. The check valve **20** may be provided at any position of the second passage **17**.
(44) The cooling system **1** includes a control device **21**. The control device **21** controls the operation of each of the first pump **18** and the second pump **19**.
(45) Specifically, when it is necessary to cool the fuel cell **2** with the coolant and it is not necessary to cool the resistor **3** with the coolant, the control device **21** operates the first pump **18** and stops the second pump **19**. As a result, the coolant discharged from the first pump **18** flows in order of the fuel cell **2** and the radiator **11**, and then is sucked into the first pump **18**. At this time, the coolant receives heat from the fuel cell **2** and radiates the heat in the radiator **11**. As a result, the fuel cell **2** is cooled.
(46) When it is necessary to cool the resistor **3** with the coolant and it is not necessary to cool the fuel cell **2** with the coolant, the control device **21** operates the second pump **19** and stops the first pump **18**. As a result, the coolant discharged from the second pump **19** flows in order of the resistor **3** and the radiator **11**, and then is sucked into the second pump **19**. At this time, the coolant receives heat from the resistor **3** and radiates the heat in the radiator **11**. Thus, the resistor **3** is cooled.
(47) When the fuel cell **2** needs to be cooled and the resistor **3** needs to be cooled, the control device **21** operates the first pump **18** and the second pump **19**. As a result, the coolant discharged from the first pump **18** flows through the fuel cell **2**, and the coolant discharged from the second pump **19** flows through the resistor **3**. The coolant flowing out of the fuel cell **2** and the coolant flowing out of the resistor **3** merge at the confluence portion **15** and flow into the radiator **11**. The coolant flowing out of the radiator **11** branches at the branch portion **14** and is sucked into each of the first pump **18** and the second pump **19**. Thus, the fuel cell **2** and the resistor **3** are cooled.
(48) Next, the effect of the cooling system **1** of the present embodiment will be described.
(49) (1) The cooling system **1** includes the radiator **11**, the outlet passage **12**, the inlet passage **13**, the branch portion **14**, the confluence portion **15**, the first passage **16**, the second passage **17**, the fuel cell **2**, the first pump **18**, the resistor **3**, and the second pump **19**.
(50) In a comparison example, if the cooling system separately has a radiator for the fuel cell for radiating the coolant received from the fuel cell **2** and a radiator for the resistor for radiating the coolant received from the resistor **3**, the cooling system becomes large.
(51) In contrast, the cooling system of the present embodiment includes one radiator **11** for the combination of the fuel cell **2** and the resistor **3**. The one radiator is shared for radiating heat of both of the coolant received from the fuel cell **2** and the coolant received from the resistor **3**. Therefore, the cooling system **1** can be downsized compared with a case where the cooling system includes a radiator for a fuel cell and a radiator for a resistor separately.
(52) Further, as a cooling system different from the present embodiment, a cooling system of Comparative Example 1 shown in FIG. **2** is be considered. The cooling system of Comparative Example 1 includes a coolant circuit **10**A in which a fuel cell **2** and a resistor **3** are connected in parallel to one radiator **11**. In the coolant circuit **10**A, only one pump **31** is connected between the radiator **11** and the fuel cell **2** and the resistor **3** connected in parallel. In the cooling system of

Comparative Example 1, it is difficult to control each of the flow rate of the coolant flowing through the fuel cell **2** and the flow rate of the coolant flowing through the resistor **3** to target values.
(53) In contrast, in the cooling system **1** of the present embodiment, the first pump **18** and the second pump **19** are provided for the fuel cell **2** and the resistor **3**, respectively. Therefore, it is easy to control each of the flow rate of the coolant flowing through the fuel cell **2** and the flow rate of the coolant flowing through the resistor **3** to the target values.
(54) (2) The cooling system **1** includes the check valve **20** provided in the second passage **17**. The first passage **16** is not provided with a check valve. The first passage **16** has no backflow preventer as a mechanical component different from the first pump **18**, which prevents the coolant from flowing in a reverse direction from the coolant outlet **2***b* to the coolant inlet **2***a*, inside of the fuel cell **2**.
(55) As a cooling system different from the present embodiment, a cooling system of Comparative Example 2 shown in FIG. **3** is considered. In the cooling system of Comparative Example 2, the fuel cell **2** and the resistor **3** are connected in parallel to one radiator **11**, and a coolant circuit **10**B is provided with a first pump **18** and a second pump **19** respectively for the fuel cell **2** and the resistor **3** as in the present embodiment. In the coolant circuit **10**B, unlike the coolant circuit **10** of the present embodiment, the check valve **20** is not provided in the second passage **17**.
(56) The frequency of cooling the resistor **3** is lower than the frequency of cooling the fuel cell **2**. Therefore, it may not be necessary to cool the resistor **3** while it is necessary to cool the fuel cell **2**. In this case, the control device **21** operates the first pump **18** and stops the second pump **19**. As a result, in the cooling system of Comparative Example 2, as shown by the arrow F**1** in FIG. **3**, a part of the coolant flowing out from the fuel cell **2** flows through the radiator **11**. The second pump **19** is a non-volumetric pump having an impeller. Therefore, as shown by the arrow F**2** in FIG. **3**, the other part of the coolant flowing out of the fuel cell **2** flows back in the second passage **17** and flows into the fuel cell **2** without flowing through the radiator **11**. As a result, the present inventors found that the heat dissipation capacity of the radiator **11** is not sufficiently exerted for cooling the fuel cell **2**.
(57) In contrast, according to the cooling system **1** of the present embodiment, the check valve **20** is provided in the second passage **17**. Therefore, it is possible to prevent the coolant flowing out of the fuel cell **2** from flowing backward through the second passage **17** when the second pump **19** is stopped and the first pump **18** is operating. All of the coolant flowing out of the fuel cell **2** can flow to the radiator **11**. As a result, the heat dissipation capacity of the radiator **11** can be sufficiently exerted for cooling the fuel cell **2**.
(58) When the resistor **3** needs to be cooled and the fuel cell **2** does not need to be cooled, the second pump **19** operates for cooling the resistor **3** and the first pump **18** stops. At this time, when a part of the coolant flowing out from the resistor **3** flows back into the first passage **16** and flows into the resistor **3** without flowing through the radiator **11**, the heat dissipation capacity of the radiator **11** is not fully exhibited for cooling the resistor **3**. Therefore, from the viewpoint of sufficiently exerting the heat dissipation capacity of the radiator **11** for cooling the resistor **3**, it is conceivable to provide a check valve in the first passage **16** in the cooling system **1** of the present embodiment. This check valve is a backflow preventer for preventing the coolant from flowing in the reverse direction.
(59) However, the amount of heat radiation required for the resistor **3** is smaller than the amount of heat radiation required for the fuel cell **2**. Therefore, even if the heat dissipation capacity of the radiator **11** is not sufficiently exerted for cooling the resistor **3**, since a part of the coolant flowing out of the resistor **3** flows back through the first passage **16**, the influence is small, compared with a case of cooling the fuel cell **2**.
(60) Rather, if a check valve is provided in the first passage **16**, the pressure loss of the coolant is increased by the check valve when the coolant flows through the first passage **16**, while the first

pump **18** is operated for cooling the fuel cell **2**. Therefore, in order to secure the same flow rate of the coolant flowing through the first passage **16** as in the case where the check valve is not provided, the rotation speed of the first pump **18** provided in the first passage **16** need to be increased. This causes issues such as a decrease in the life of the first pump **18** and a deterioration in the fuel efficiency. Further, by providing the check valve in the first passage **16**, the number of components needed for the cooling system is increased. Since the cooling system is mounted on the vehicle, it is better to have few components of the cooling system. These are the same even when a backflow preventer other than the check valve is provided in the first passage **16**.
(61) Therefore, according to the cooling system **1** of the present embodiment, the first passage **16** is not provided with a backflow preventer such as check valve. As a result, the pressure loss generated in the coolant flowing through the first passage **16** can be reduced as compared with a case where the check valve or the like is provided in the first passage **16**. Further, the number of components needed for the cooling system **1** can be reduced as compared with a case where a check valve or the like is provided in both the first passage **16** and the second passage **17**.
Second Embodiment
(62) As shown in FIG. **4**, the present embodiment differs from the first embodiment in that the cooling system **1** includes a flow meter **22**. Further, the present embodiment is different from the first embodiment in that the control device **21** controls the operation of the first pump **18** based on the measurement result of the flow meter **22** when the second pump **19** is operating. Other configurations of the cooling system **1** are the same as those of the first embodiment.
(63) The flow meter **22** measures the flow rate of the coolant flowing through the first passage **16**. The flow meter **22** may be of any measurement type such as an electromagnetic type and an ultrasonic type. The flow meter **22** is provided at a position in the first passage **16** between the fuel cell **2** and the confluence portion **15**. The position where the flow meter **22** is provided may be between the first pump **18** and the fuel cell **2** in the first passage **16**. Further, the position where the flow meter **22** is provided may be between the branch portion **14** and the first pump **18** in the first passage **16**.
(64) The flow meter **22** is connected to the input side of the control device **21**. The control device **21** can acquire the measurement result of the flow meter **22**. The control device **21** controls the rotation speed of the first pump **18**. By controlling the rotation speed of the first pump **18** by the control device **21**, the discharge capacity of the first pump **18** is adjusted.
(65) When it is necessary to cool the resistor **3** with the coolant and it is not necessary to cool the fuel cell **2** with the coolant, the control device **21** operates the second pump **19** and operates the first pump **18**. At this time, the control device **21** operates the second pump **19** at a predetermined rotation speed so that the flow rate of the coolant flowing through the second passage **17** becomes the flow rate required for cooling the resistor **3**. The control device **21** operates the first pump **18** at a rotation speed lower than a rotation speed of the first pump **18** necessary for cooling the fuel cell **2**.
(66) Further, the control device **21** controls the rotation speed of the first pump **18** based on the measurement result of the flow meter **22** so that the coolant flows in the first passage **16** in the forward direction. The forward direction is a direction in which the coolant flows from the coolant inlet **2***a* to the coolant outlet **2***b* inside of the fuel cell **2**.
(67) Specifically, when the second pump **19** is operating, the control device **21** drives the first pump **18** and repeatedly performs the control process shown in FIG. **5**. At the start of operation of the first pump **18**, the rotation speed of the first pump **18** is set to a predetermined rotation speed. The steps shown in FIG. **5** correspond to functional units that realize various functions. This also applies to other figures.
(68) In step S**1**, the control device **21** acquires the flow rate of the coolant flowing through the first passage **16** measured by the flow meter **22**.
(69) Subsequently, in step S**2**, the control device **21** compares the acquired flow rate with a

predetermined value, and determines whether the acquired flow rate is smaller than the predetermined value. When the flow direction of the coolant in the first passage **16** is in the forward direction, the flow rate measured by the flow meter **22** is a positive value larger than 0. When the flow direction of the coolant in the first passage **16** is reversed, the flow rate measured by the flow meter **22** is a negative value smaller than 0. Therefore, the predetermined value is set to 0 or a positive value so that the coolant in the first passage **16** flows in the forward direction.

(70) When the determination is YES in step S**2**, the control device **21** proceeds to step S**3** and increases the rotation speed of the first pump **18**. After that, the control device **21** temporarily ends the control process shown in FIG. **5**. After that, the control device **21** again performs the control process shown in FIG. **5**. On the other hand, when the NO determination is made in step S**2**, the control device **21** temporarily ends the control process shown in FIG. **5**. After that, the control device **21** again performs the control process shown in FIG. **5**.

(71) Unlike the present embodiment, when the resistor **3** needs to be cooled and the fuel cell **2** does not need to be cooled, the control device **21** drives the second pump **19** to cool the resistor **3**, and stops the first pump **18**. At this time, a part of the coolant flowing out from the resistor **3** may flow back in the first passage **16**.

(72) The backflow of the coolant through the first passage **16** leads to a failure of the first pump **18**. Further, in case where a filter for preventing the intrusion of a foreign matter is installed only in the coolant inlet **2***a* but not provided in the coolant outlet **2***b* of the fuel cell **2**, if the coolant flows back through the first passage **16**, the foreign matter cannot be prevented from entering the fuel cell **2**. For these reasons, it is preferable that the coolant flows in the first passage **16** in the forward direction.

(73) Therefore, in the present embodiment, as described in steps S**2** and S**3**, the control device **21** increases the rotation speed of the first pump **18** when the flow rate measured by the flow meter **22** is smaller than a predetermined value. In this way, the control device **21** ensures that the flow rate of the coolant flowing through the first passage **16** does not fall below the predetermined value, that is, the flow rate of the coolant flowing through the first passage **16** becomes larger than the predetermined value, by controlling the rotation speed of the first pump **18**. As a result, the coolant can flow in the forward direction in the first passage **16**. By setting the rotation speed of the first pump **18** to be lower than the rotation speed of the first pump **18** necessary to cool the fuel cell **2**, the coolant flows through the first passage **16** at a flow rate lower than the flow rate of the coolant necessary to cool the fuel cell **2**.

(74) Although the flow rate is acquired as the measurement result of the flow meter **22** in step S**1**, the output value of the flow meter **22** before being converted into the flow rate value may be acquired as the measurement result of the flow meter **22**. In this case, the predetermined value used in step S**2** is set to a value corresponding to the output value of the flow meter **22** when the flow rate of the coolant flowing through the first passage **16** is 0 or the positive value.

Third Embodiment

(75) As shown in FIG. **6**, the cooling system **1** of the present embodiment includes a current meter **23** instead of the flow meter **22** of the second embodiment. The control device **21** performs the control process shown in FIG. **7** instead of the control process shown in FIG. **5** of the second embodiment. Other than these, the third embodiment is the same as the second embodiment.

(76) The current meter **23** measures the flow velocity of the coolant flowing through the first passage **16**. The current meter **23** may be of any measurement type such as an electromagnetic type and a laser Doppler type. The current meter **23** is provided at a position between the fuel cell **2** and the confluence portion **15** in the first passage **16**. The position where the current meter **23** is provided may be between the first pump **18** and the fuel cell **2** in the first passage **16**. Further, the position where the current meter **23** is provided may be between the branch portion **14** and the first pump **18** in the first passage **16**. The current meter **23** is connected to the input side of the control device **21**. The control device **21** can acquire the measurement result of the current meter **23**.

(77) The control device **21** drives the first pump **18** when the second pump **19** is operating, and controls the rotation speed of the first pump **18** based on the measurement result of the current meter **23** so that the coolant flows in the first passage **16** in the forward direction. Specifically, when the second pump **19** is operating, the control device **21** operates the first pump **18** and repeatedly performs the control process shown in FIG. **7**.

(78) As shown in FIG. **7**, in step S**11**, the control device **21** acquires the flow velocity of the coolant flowing through the first passage **16** measured by the current meter **23**.

(79) Subsequently, in step S**12**, the control device **21** compares the acquired flow velocity with a predetermined value, and determines whether the acquired flow velocity is smaller than the predetermined value. When the flow direction of the coolant in the first passage **16** is in the forward direction, the flow velocity measured by the current meter **23** is a positive value larger than 0. When the flow direction of the coolant in the first passage **16** is reversed, the flow velocity measured by the current meter **23** has a negative value smaller than 0. When there is no flow of the coolant in the first passage **16**, the flow velocity measured by the current meter **23** becomes 0. Therefore, the predetermined value is set to 0 or a positive value so that the coolant in the first passage **16** flows in the forward direction.

(80) When the determination is YES in step S**12**, the control device **21** proceeds to step S**13** and increases the rotation speed of the first pump **18**. After that, the control device **21** temporarily ends the control process shown in FIG. **7**. After that, the control device **21** again performs the control process shown in FIG. **7**. On the other hand, when the NO determination is made in step S**12**, the control device **21** temporarily ends the control process shown in FIG. **7**. After that, the control device **21** again performs the control process shown in FIG. **7**.

(81) As described in steps S**12** and S**13**, the control device **21** increases the rotation speed of the first pump **18** when the flow velocity measured by the current meter **23** is smaller than a predetermined value. In this way, the control device **21** ensures that the flow velocity of the coolant flowing through the first passage **16** does not fall below the predetermined value, that is, the flow velocity of the coolant flowing through the first passage **16** becomes larger than the predetermined value, by controlling the rotation speed of the first pump **18**. As a result, the same effect as that of the second embodiment can be obtained.

Fourth Embodiment

(82) As shown in FIG. **8**, the cooling system **1** of the present embodiment includes a first pressure sensor **24** and a second pressure sensor **25** in place of the flow meter **22** of the second embodiment. The control device **21** performs the control process shown in FIG. **9** instead of the control process shown in FIG. **5** of the second embodiment. Other than these, the fourth embodiment is the same as the second embodiment.

(83) The first pressure sensor **24** and the second pressure sensor **25** measure the pressure of the coolant at two distant points in the first passage **16**. The first pressure sensor **24** and the second pressure sensor **25** are provided at two locations on the downstream side of the outlet of the first pump **18** in a flow of the coolant in the first passage **16**. Specifically, the first pressure sensor **24** is provided at a position between the outlet of the first pump **18** and the coolant inlet **2***a* of the fuel cell **2** in the first passage **16**. The second pressure sensor **25** is provided at a position between the coolant outlet **2***b* of the fuel cell **2** and the confluence portion **15** in the first passage **16**. In this way, the second pressure sensor **25** is arranged away from the first pressure sensor **24** on the downstream side in a flow of the coolant when the coolant flows in the forward direction through the first passage **16**.

(84) The first pressure sensor **24** and the second pressure sensor **25** are connected to the input side of the control device **21**. The control device **21** can acquire the measurement results of the first pressure sensor **24** and the second pressure sensor **25**.

(85) When the second pump **19** is operating, the control device **21** drives the first pump **18** and controls the rotation speed of the first pump **18** based on the measurement result of the first

pressure sensor **24** and the second pressure sensor **25**. Specifically, the control device **21** controls the rotation speed of the first pump **18** based on a relationship between the magnitude relationship of the first pressure value and the second pressure value, and the flow direction of the coolant in the first passage **16**, such that the coolant flows in the forward direction in the first passage **16**. Specifically, when the second pump **19** is operating, the control device **21** operates the first pump **18** and repeatedly performs the control process shown in FIG. **9**.
(86) As shown in FIG. **9**, in step S**21**, the control device **21** acquires the first pressure value measured by the first pressure sensor **24** and the second pressure value measured by the second pressure sensor **25**.
(87) Subsequently, in step S**22**, the control device **21** compares the first pressure value and the second pressure value acquired in step S**21**, and determines whether the first pressure value is smaller than the second pressure value. Here, when the flow direction of the coolant in the first passage **16** is in the forward direction, the first pressure value is larger than the second pressure value. When the flow direction of the coolant in the first passage **16** is reversed, the first pressure value is smaller than the second pressure value. These are the relationship between the magnitude relationship of the first pressure value and the second pressure value and the flow direction of the coolant in the first passage **16**.
(88) When the determination is YES in step S**22**, the control device **21** proceeds to step S**23** and increases the rotation speed of the first pump **18**. After that, the control device **21** temporarily ends the control process shown in FIG. **9**. After that, the control device **21** again performs the control process shown in FIG. **9**. On the other hand, when the NO determination is made in step S**22**, the control device **21** temporarily ends the control process shown in FIG. **9**. After that, the control device **21** again performs the control process shown in FIG. **9**.
(89) As described in steps S**22** and S**23**, the control device **21** increases the rotation speed of the first pump **18** when the first pressure value is smaller than the second pressure value. In this way, the control device **21** controls the rotation speed of the first pump **18** so that the first pressure value becomes larger than the second pressure value. As a result, the same effect as that of the second embodiment can be obtained.
(90) The positions where the first pressure sensor **24** and the second pressure sensor **25** are provided may be other positions while being located at two locations on the downstream side of the outlet of the first pump **18** in a flow of the coolant in the first passage **16**. Further, the positions where the first pressure sensor **24** and the second pressure sensor **25** are provided may be any two positions on the upstream side of the inlet of the first pump **18** in a flow of the coolant in the first passage **16**.
Fifth Embodiment
(91) As shown in FIG. **10**, the cooling system **1** of the present embodiment includes a differential pressure sensor **26** instead of the flow meter **22** of the second embodiment. The control device **21** performs the control process shown in FIG. **11** instead of the control process shown in FIG. **5** of the second embodiment. Other than these, the fifth embodiment is the same as the second embodiment.
(92) The differential pressure sensor **26** measures the difference in pressure between the coolants at two distant locations in the first passage **16**, that is, the differential pressure. The differential pressure sensor **26** is provided at a position between the coolant outlet **2***b* of the fuel cell **2** and the confluence portion **15** in the first passage **16**. The position where the flow meter **22** is provided may be provided at another position in the first passage **16**.
(93) The differential pressure sensor **26** is connected to the input side of the control device **21**. The control device **21** can acquire the measurement result of the differential pressure sensor **26**. The control device **21** drives the first pump **18**, when the second pump **19** is operating, and controls the rotation speed of the first pump **18** based on the measurement result of the differential pressure sensor **26**, specifically, a relationship between the magnitude of the differential pressure and the flow direction of the coolant in the first passage **16**, so that the coolant flows in the forward

direction in the first passage **16**. Specifically, when the second pump **19** is operating, the control device **21** drives the first pump **18** and repeatedly performs the control process shown in FIG. **11**.
(94) As shown in FIG. **11**, in step S**31**, the control device **21** acquires the differential pressure value measured by the differential pressure sensor **26**.
(95) Subsequently, in step S**32**, the control device **21** compares the differential pressure value acquired in step S**21** with a predetermined value, and determines whether the differential pressure value is smaller than the predetermined value. Here, when the flow direction of the coolant in the first passage **16** is in the forward direction, the differential pressure value measured by the differential pressure sensor **26** is a positive value larger than 0. When the flow direction of the coolant in the first passage **16** is reversed, the differential pressure value measured by the differential pressure sensor **26** is a negative value smaller than 0. These are the relationship between the magnitude of the differential pressure and the flow direction of the coolant in the first passage **16**. Therefore, the predetermined value is set to 0 or a positive value so that the coolant in the first passage **16** flows in the forward direction.
(96) When the determination is YES in step S**32**, the control device **21** proceeds to step S**33** and increases the rotation speed of the first pump **18**. After that, the control device **21** temporarily ends the control process shown in FIG. **11**. After that, the control device **21** again performs the control process shown in FIG. **11**. On the other hand, when the NO determination is made in step S**22**, the control device **21** temporarily ends the control process shown in FIG. **11**. After that, the control device **21** again performs the control process shown in FIG. **11**.
(97) As described in steps S**32** and S**33**, the control device **21** increases the rotation speed of the first pump **18** when the differential pressure value measured by the differential pressure sensor **26** is smaller than a predetermined value. In this way, the control device **21** prevents the differential pressure value of the coolant flowing through the first passage **16** from falling below the predetermined value, that is, the differential pressure value of the coolant flowing through the first passage **16** is greater than the predetermined value, by controlling the rotation speed of the first pump **18**. As a result, the same effect as that of the second embodiment can be obtained.
Sixth Embodiment
(98) As shown in FIG. **12**, the present embodiment differs from the first embodiment in that the cooling system **1** includes an intercooler **4**, a bypass passage **27**, and a distribution valve **28**. Other configurations of the cooling system **1** are the same as those of the first embodiment.
(99) The intercooler **4** cools air pumped from an air pump (not shown) to the fuel cell **2** with a coolant. The air pumped by the air pump is compressed and warmed up. The intercooler **4** is connected in parallel with the fuel cell **2** between the first pump **18** and the confluence portion **15** in the first passage **16**.
(100) More specifically, the first passage **16** includes a third passage **161** and a fourth passage **162** arranged in parallel between the first pump **18** and the confluence portion **15**. The fuel cell **2** is arranged in the middle of the third passage **161**. The intercooler **4** is arranged in the middle of the fourth passage **162**.
(101) The bypass passage **27** bypasses the radiator **11** to allow the coolant flowing out of the fuel cell **2** to bypass the radiator **11**. One end of the bypass passage **27** is connected in the middle of the inlet passage **13**. The other end of the bypass passage **27** is connected in the middle of the outlet passage **12**.
(102) The distribution valve **28** distributes the coolant flowing out of the fuel cell **2** to the radiator **11** and the bypass passage **27**, and controls a flow rate of the coolant flowing through the radiator **11** and a flow rate of the coolant flowing through the bypass passage **27**. The distribution valve **28** is provided at a connection portion between the one end of the bypass passage **27** and the inlet passage **13**.
(103) The distribution valve **28** is electrically connected to the output side of the control device **21**. The control device **21** controls the operation of the distribution valve **28** for adjusting the

temperature of the fuel cell **2**. By controlling the operation of the distribution valve **28**, the flow rate of the coolant flowing through the radiator **11** and the flow rate of the coolant flowing through the bypass passage **27** are adjusted. The temperature of the fuel cell **2** is adjusted to an appropriate temperature.

(104) The same effect as that of the first embodiment can be obtained by the same configuration as that of the first embodiment also in this embodiment. Further, the second to fifth embodiments can be applied to the present embodiment. As a result, the same effect as that of the second to fifth embodiments can be obtained.

Other Embodiments

(105) (1) In each of the embodiments, the check valve **20** is used as a backflow preventer. Alternatively, an on-off valve may be used as a backflow preventer. In this case, when the second pump **19** is stopped, the on-off valve is closed. As a result, the backflow of the coolant in the second passage **17** is prevented when the second pump **19** is stopped and the first pump **18** is operated.

(106) (2) In each of the embodiments, the radiator **11** exchanges heat between the coolant and air. Alternatively, a heat exchanger that exchanges heat between the coolant and a refrigerant of a refrigeration cycle may be used as the radiator.

(107) (3) In each of the embodiments, the first pump **18** is provided on the upstream side of the fuel cell **2** in a flow of the coolant in the first passage **16**. However, the first pump **18** may be provided on the downstream side of the fuel cell **2** in a flow of the coolant in the first passage **16**. Similarly, the second pump **19** may be provided on the downstream side of the resistor **3** in a flow of the coolant in the second passage **17**.

(108) (4) The cooling system of each of the embodiments is mounted on a large commercial vehicle equipped with the fuel cell **2**. However, the cooling system may be mounted on a passenger car equipped with the fuel cell **2**. Further, the cooling system is not limited to be mounted on the passenger car and the commercial vehicle, and may be mounted on other vehicles having wheels and traveling on land, such as railroad vehicles, industrial vehicles used for cargo handling and transportation, construction vehicles used for construction, agricultural vehicles used for agriculture, and the like. Further, the electric power generated by the fuel cell **2** may be used for other than driving the motor generator mounted on the vehicle.

(109) (5) The present disclosure is not limited to the foregoing description of the embodiments and can be modified within the scope of the present disclosure. The present disclosure may also be varied in many ways. Such variations are not to be regarded as departure from the disclosure, and all such modifications are intended to be included within the scope of the disclosure. The embodiments described above are not independent of each other, and can be appropriately combined except when the combination is obviously impossible. Further, in each of the embodiments, it goes without saying that components of the embodiment are not necessarily essential except for a case in which the components are particularly clearly specified as essential components, a case in which the components are clearly considered in principle as essential components, and the like.

(110) (6) The control device and the technique according to the present disclosure may be achieved by a dedicated computer provided by constituting a processor and a memory programmed to execute one or more functions embodied by a computer program. Alternatively, the control device and the technique according to the present disclosure may be achieved by a dedicated computer provided by constituting a processor with one or more dedicated hardware logic circuits. Alternatively, the control device and the method described in the present disclosure may be implemented by one or more special purpose computer, which is configured as a combination of a processor and a memory, which are programmed to perform one or more functions, and a processor which is configured with one or more hardware logic circuits. The computer program may be

stored in a computer-readable non-transitory tangible storage medium as an instruction executed by a computer.

Claims

1. A cooling system configured to cool a fuel cell for a vehicle comprising: a radiator configured to radiate heat of a coolant, the radiator having a radiator inlet for a coolant to flow in and a radiator outlet for a coolant to flow out; an outlet passage connected to the radiator outlet for the coolant to flow out of the radiator outlet; an inlet passage connected to the radiator inlet for the coolant to flow in the radiator inlet; a branch portion provided in the outlet passage for branching into a plurality of passages for a coolant; a confluence portion provided in the inlet passage for merging the plurality of passages into one passage; a first passage and a second passage, as the plurality of passages, connected in parallel with each other between the branch portion and the confluence portion, a coolant flowing from the branch portion toward the confluence portion; a fuel cell provided in the first passage, the fuel cell having a fuel cell inlet for a coolant to flow in and a fuel cell outlet for a coolant to flow out, a coolant flowing inside the fuel cell, the fuel cell generating electric power by an electrochemical reaction of fuel; a first pump provided in the first passage to pump a coolant; a resistor provided in the second passage for a brake, the resistor having a resistor inlet for a coolant to flow in and a resistor outlet for a coolant to flow out, a coolant flowing inside the resistor, the resistor converting an electric power generated by a motor generator into a heat energy by a braking operation to rotation of a wheel mechanically connected to the motor generator; a second pump provided in the second passage to pump a coolant; and a backflow preventer provided in the second passage as a mechanical component different from the second pump, so as to prevent a coolant from flowing inside the resistor in a reverse direction from the resistor outlet to the resistor inlet, wherein the first passage has no backflow preventer, as a mechanical component different from the first pump, which prevents a coolant from flowing inside the fuel cell in a reverse direction from the fuel cell outlet to the fuel cell inlet.
2. The cooling system according to claim 1, further comprising: a control device configured to control the first pump and the second pump, wherein the control device drives the first pump when the second pump is operating, and controls a rotation speed of the first pump such that a coolant flows through the first passage in a forward direction from the fuel cell inlet to the fuel cell outlet inside the fuel cell.